

Intermediate Macroeconomics: Lecture W4 (Ch.11)

ECON 311 – Intermediate Macroeconomics

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Fall 2026

What are You Going to Learn (Ch. 11)

- The IS curve
 - The foundation component of the short-run model
 - Illustrates the negative correlation between the real interest rate and short-run output
- Understanding how shocks to consumption, investment, government purchases, or net exports—referred to as "aggregate demand shocks" – can alter the position of the IS curve
- Highlighting investment as the primary channel through which fluctuations in real interest rates impact GDP in the short run
- The life-cycle/permanent-income hypothesis can form the microfoundations for the IS curve

Introduction to the IS Curve

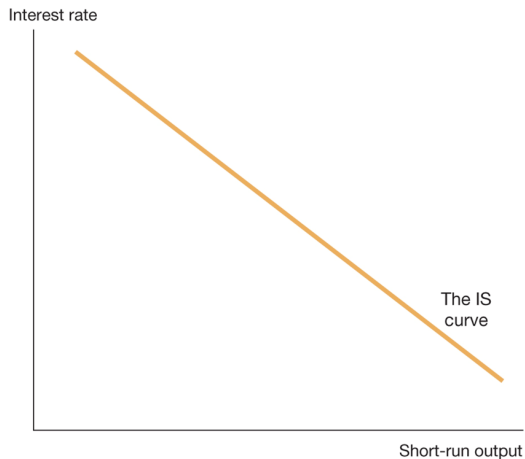
- The basic story is as follows:

$$\uparrow \text{Interest Rate} \implies \downarrow \text{Investment} \implies \downarrow \text{Output}$$

- The same story can apply to college students, individuals taking out loans, financial advisors, firms and even people who do not borrow money
- The IS curve illustrates the negative relationship between interest rates and short-run output

Introduction to the IS Curve

- IS curve captures how a high interest rate reduces output in the short run (and vice versa) through which this occurs is investment



Setting Up the Economy

- The short-run model, just like the long-run model of economic growth, is based on a simple economy and various assumptions
- First, define how resource are spent in an economy
- The national income accounting identity
 - total resources available to the economy equal total uses

$$Y_t = C_t + I_t + G_t + EX_t - IM_t$$

where

C_t is the consumption

Y_t is the output

I_t is the investment

G_t is the government spending

$EX_t - IM_t$ is the net export

Setting Up the Economy

Five additional equations to solve the model

$$C_t = \bar{a}_c \bar{Y}_t$$

$$G_t = \bar{a}_g \bar{Y}_t$$

$$EX_t = \bar{a}_{ex} \bar{Y}_t$$

$$IM_t = \bar{a}_{im} \bar{Y}_t$$

$$\frac{I_t}{\bar{Y}_t} = \bar{a}_i - \bar{b}(R_t - \bar{r})$$

- Except for investment, all component are a proportion of the potential output $\bar{Y}_t$
- Investment is determined by an exogenous part and is depends on fluctuations in the interest rate from its long-run level, $\bar{r}$

Consumption, Government Purchases, Net Export

- Empirically $\bar{a}_c$ in the consumption equation estimated to be approximately 2/3
 - two out of every three dollars of GDP is attributed to consumption
 - as suggested by the permanent income hypothesis, the share is relatively stable
- Furthermore, we set government purchases, exports, and imports are all assumed to be long-run fractions of output as see in the data
 - The other variables can change very often, but for simplicity, we will assume them to be constant for now

The Investment Equation

$$\frac{I_t}{\bar{Y}_t} = \bar{a}_i - \bar{b}(R_t - \bar{r})$$

- Part of investment is determined by a constant fraction $\bar{a}_i$ of potential output, $\bar{Y}$
- The amount of investment also depends on the gap between the real interest rate R_t and the marginal product of capital ($\bar{r}$)
 - The effect of the difference would be depend on the parameter $\bar{b}$
- The long-run interest rate can be interpret as the Marginal Product of Capital (MPK)

Marginal Product of Capital (MPK)

- MPK: Amount of additional output the firm can produce by investing in one more unit of capital
- Here, the MPK is determined in the long-run model, which is exogenous and time invariant in this short-run model

Marginal Product of Capital (MPK)

- In the short run, the gap between MPK and the real interest rate determine investment
 - If $R_t > MPK$, firm/capital owner is better off to invest in financial market (saving), investment falls
 - If $R_t < MPK$, firm/capital owner is better off to invest in firm (or renting capital), investment rises
 - In the long run, all these adjustment would be made and thus $R_t = \bar{r} = MPK$
- Why would R_t be different from $\bar{r}$
 - Capital takes time to build, Central bank's policy, etc

Potential Output

- In the long run, $R_t - \bar{r}$, the investment is then

$$I_t = \bar{a}_i \bar{Y}$$

- Summing all the equations, we have

$$\begin{aligned} Y_t &= C_t + G_t + EX_t - IM_t + I = \bar{a}_c \bar{Y} + \bar{a}_g \bar{Y} + \bar{a}_{ex} \bar{Y} - \bar{a}_{im} \bar{Y} + \bar{a}_i \bar{Y} \\ &= (\bar{a}_c + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} + \bar{a}_i) \bar{Y} \end{aligned}$$

- Then, together with $(\bar{a}_c + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} + \bar{a}_i) = 1$ implies the economy is in the long run, which $Y_t = \bar{Y}$
- This would also imply that the only source causing the output be deviate from $\bar{Y}$ is $R_t - \bar{r}$ and/or when $(\bar{a}_c + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} + \bar{a}_i) \neq 1$ in this simple model

The Setup of the Economy for the IS Curve

$$C_t = \bar{a}_c \bar{Y}_t$$

$$G_t = \bar{a}_g \bar{Y}_t$$

$$EX_t = \bar{a}_{ex} \bar{Y}_t$$

$$IM_t = \bar{a}_{im} \bar{Y}_t$$

$$\frac{I_t}{\bar{Y}_t} = \bar{a}_i - \bar{b}(R_t - \bar{r})$$

$$Y_t = C_t + I_t + G_t + EX_t - IM_t$$

- Six equations
- Endogenous variables: $Y_t, C_t, I_t, G_t, EX_t, IM_t$
- Exogenous variables/parameters: $\bar{Y}, \bar{r}, \bar{a}_c, \bar{a}_g, \bar{a}_{ex}, \bar{a}_{im}, \bar{a}_i, \bar{b}, R_t$

Deriving the IS Curve

- Recall the definition of short-run output

$$\tilde{Y}_t = \frac{Y_t - \bar{Y}}{\bar{Y}} = \frac{Y_t}{\bar{Y}} - 1$$

- Sub in $Y_t = (\bar{a}_c + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} + \bar{a}_i - \bar{b}(R_t - \bar{r})) \bar{Y}$

$$\frac{Y_t}{\bar{Y}} - 1 = \bar{a}_c + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} + \bar{a}_i - \bar{b}(R_t - \bar{r}) - 1$$

- Let $\bar{a} = \bar{a}_c + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} + \bar{a}_i - 1$,

$$\tilde{Y}_t = \frac{Y_t}{\bar{Y}} - 1 = \bar{a} - \bar{b}(R_t - \bar{r})$$

Deriving the IS Curve

- Variable C , G , XM and IM can be deviated from their long-run level in the short run,

$$\bar{a} = \bar{a}_c + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} + \bar{a}_i - 1 \neq 0$$

- It can be interpret as aggregate demand shock
- Rewrite the formula above

$$\tilde{Y} = \bar{a} - \bar{b}(R_t - \bar{r})$$

- In the long run, $\tilde{Y} = 0$ and $\bar{a} = 0$

$$R_t = \bar{r}$$

Why Is It Called the "IS Curve"?

- The concept was introduced by John R. Hicks
- "IS" stands for "investment equals savings."

$$Y = C + I + G + EX - IM$$

$$Y - C - G + (IM - EX) = I$$

- Add and subtract T , government tax

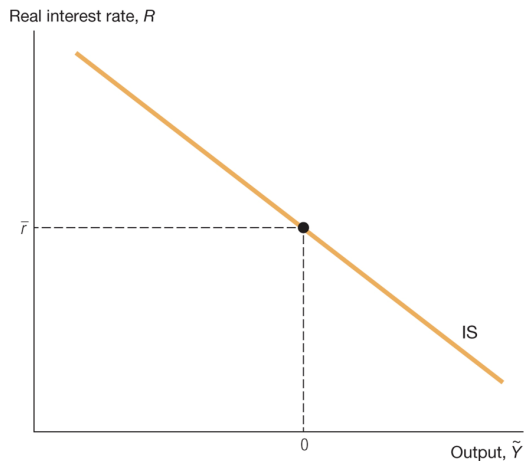
$$(Y - T - C) + (T - G) + (IM - EX) = I$$

- $Y - T - C$ is private saving
- $T - G$ is the public saving
- $IM - EX$ is foreign saving

Using the IS Curve

$$\tilde{Y} = \bar{a} - \bar{b}(R_t - \bar{r}) \implies R_t = \bar{r} + \frac{\bar{a}}{\bar{b}} - \frac{\tilde{Y}}{\bar{b}}$$

In the long run, $\bar{a} = 0$, $\tilde{Y} = 0$



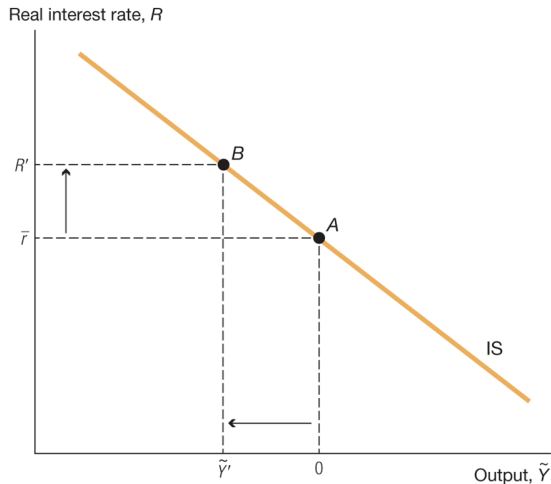
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A Change in the Interest Rate

- A change in the real interest rate moves the economy along the IS curve
 1. Increase in interest rate (exogenous)
 2. Increase in borrowing costs/opportunity cost of investing
 3. Decrease in demand for investment
 4. Decrease in output (Y_t)
 5. Move up along the IS curve

An Increase in the Real Interest Rate to R'

- An increase to R' in the real interest rate is a movement along the IS curve that moves the economy from A to B, resulting in a decline in output in the short run



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A Change in the Interest Rate

- How much would the output change as the interest rate increase?
 - It depends on $\bar{b}$

$$R_t = \frac{\bar{a}}{\bar{b}} + \bar{r} - \frac{1}{\bar{b}} \tilde{Y}$$

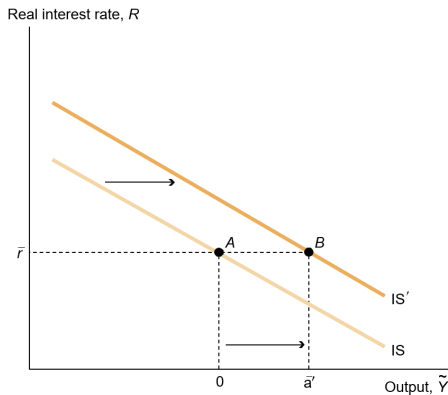
- The IS curve would be flatter as the slope $\left(\frac{1}{\bar{b}}\right)$ decreases
- The same change in the interest rate would be associated with larger changes in output

An Aggregate Demand Shock

- Suppose that there information technology boom create an investment boom
 1. Increase in optimism
 2. Increase demand for capital
 3. Increase in $\bar{a}_i$, which increase $\bar{a}$
 4. Increase investment at a given interest rate R_t
 5. IS curve shift to the right

An Aggregate Demand Shock

- The effect of a positive aggregate demand shock that raises $\bar{a}$ from 0 to $\bar{a}'$
- IS curve shifts out, moving the economy from point A to point B
 - Holding $R = \bar{r}$, the output $\tilde{Y}$ move from 0 to $\bar{a}'$
- Aggregate demand shocks typically result in one-for-one changes in short-run output



A Shock to Potential Output

- Suppose there is a discovery of a new technology (one possibility)
 1. Increase in potential output
 2. Increase in actual output
 3. Short-run output unchanged
- Recall $\tilde{Y} = \frac{Y_t}{\bar{Y}_t} - 1$. If the economy start from long-run equilibrium, if Y_t and $\bar{Y}_t$ increase by the same amount, the ratio $Y_t/\bar{Y}_t$ will equal to 1 and $\tilde{Y} = 0$
- Some shock to potential output may change other parameters
 - For example, an earthquake will reduce actual and potential output by the same amount and also may change the MPK as capital stock was destroyed

Microfoundations of the IS Curve

- Microfoundations: Explain the microeconomic behavior that establishes the demands for C , I , G , EX and IM
- In modern Macroeconomics, Microfoundations serves as the fundamental building blocks for macroeconomic models, ensuring that aggregate economic phenomena are derived from individual-level decisions
- That is because aggregating individual behaviors to derive macroeconomic variables, ensuring that the macro-level outcomes are consistent with micro-level actions
- Macro-economist model consumption and investment of the aggregate economy much like in the consumption model and investment model (in ch. 16/17)

Microfoundations of the IS Curve

- There are often individual deviations from the aggregate macroeconomic behavior. However, when we aggregate, the outliers tend to average out
 - For instance, a particular individual may spend all (or most) of his or her income on basketball tickets
 - Another individual may spend none (or almost none) of his or her income and save the remainder
 - These idiosyncratic behaviors are not part of our model, and average out when we study the economy as a whole

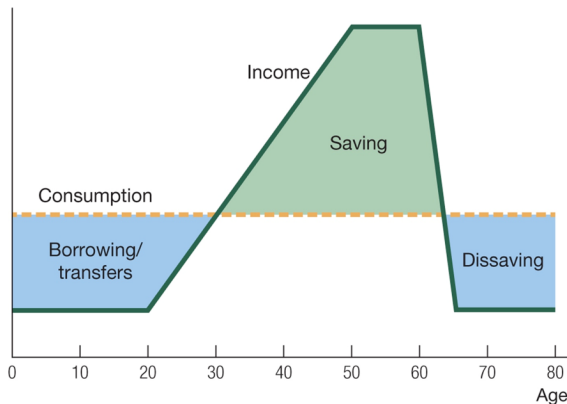
Individuals prefer to smooth their consumption spending over time

- Two hypothesis guide how consumption path is modeled: permanent-income hypothesis and life-cycle model
 - Consumption path is how much an individual consume in each period overtime
- The life-cycle/permanent-income (LC/PI) hypothesis:
 - Implies that people smooth their consumption relative to their lifetime income
 - individuals prefer a small variance in their consumption spending
 - This is why we set consumption proportional to potential output rather than actual output (i.e., $C_t = \bar{a}_c \bar{Y}_t$)

- Suppose a consumer are given the choice between Option A and Option B
- Option A:
 - Consumption: one piece of cake (\$3) every day Monday through Friday
 - Income: \$15 on Friday, but you may borrow \$3 a day (at no interest) to consume during the week
- Option B:
 - Consumption: five pieces of cake (\$15) on Friday
 - Income: \$15 on Friday
- The permanent income hypothesis predicts that people will choose Option A

The Life-Cycle Model of Consumption

- According to this theory, individuals compensate for this income variation by saving and borrowing in order to consume
- People consume more than income, by borrowing or intra-family transfer, start saving as income increase, and dis-save when retire



Empirical Evidence (Hsieh, 2003)

- How do consumers respond to two different types of income shocks in Alaska?
 - Annual payment from the State of Alaska's Permanent Fund (from oil revenues)
 - Large payment relative to income, easy to predict, the amount announce 6 mths earlier
 - Federal tax refunds
 - Smaller payment, harder to predict
- In theory, the LC/PI hypothesis predicts that consumption should not change when the Permanent Fund check or Federal tax refunds are received

- Hsieh finds the sample smooth their consumption for the Permanent Fund
- But the same is not true in response to a tax refund. For every \$1 of a tax refund, consumers spend 0.30 cents (contradictory to the predictions of the LC/PI hypothesis)
- Hsieh interprets this evidence as suggesting that the LC/PI hypothesis works well for large and easy-to-predict changes in income but less well for small and harder-to-predict shocks

Multiplier Effects

- Most evidence would suggest that household would not be smoothing consumption perfectly
- Let us to augment the consumption equation to include a term that responds to changes in short-run output
 - This yields a multiplier effect
- Consumption equal:

$$\frac{C_t}{\bar{Y}_t} = \bar{a}_c + \bar{x} \tilde{Y}_t$$

where $\bar{x} \in (0, 1)$ is a parameter that determines how much consumption rises when the economy expands

Multiplier Effects

- Solving for the IS curve with this augmented consumption function

$$Y_t = C_t + I_t + G_t + EX_t - IM_t = (\bar{a}_c + \bar{x} \tilde{Y}_t) \bar{Y} + (\bar{a}_i - \bar{b}(R_t - \bar{r})) \bar{Y} + \bar{a}_g \bar{Y} + \bar{a}_{ex} \bar{Y} - \bar{a}_{im} \bar{Y}$$

- Divide it by $\bar{Y}$

$$\frac{Y_t}{\bar{Y}} = \bar{a}_c + \bar{x} \tilde{Y}_t + \bar{a}_i - \bar{b}(R_t - \bar{r}) + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im}$$

- Recall that $\bar{a} = \bar{a}_c + \bar{a}_i + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} - 1$

$$\tilde{Y} = \frac{Y_t}{\bar{Y}} - 1 = \bar{x} \tilde{Y}_t - \bar{b}(R_t - \bar{r}) + (\bar{a}_c + \bar{a}_i + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} - 1)$$

$$\tilde{Y} = \bar{x} \tilde{Y} + \bar{a} - \bar{b}(R_t - \bar{r})$$

Multiplier Effects

- Finally, rearrange:

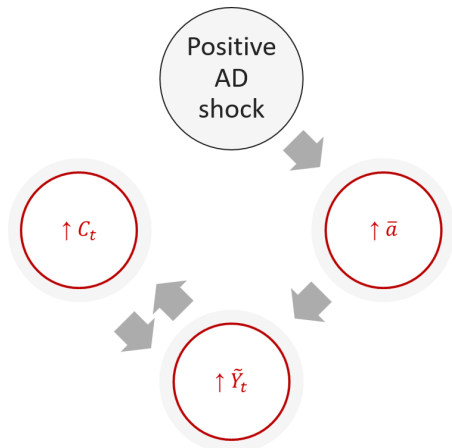
$$\tilde{Y} = \frac{1}{1 - \bar{x}} \times (\bar{a} - \bar{b}(R_t - \bar{r}))$$

- The second term $(\bar{a} - \bar{b}(R_t - \bar{r}))$ is just the original IS curve
- The first term $\frac{1}{1 - \bar{x}}$ is the multiplier
 - Given $\bar{x} \in (0, 1)$, the multiplier is greater than 1
 - Any changes in the original IS curve (i.e., $\bar{a}$, $\bar{b}$, R_t or $\bar{r}$) would be multiplied
 - Say, $\bar{a}$ increased by 1 percent and $\bar{x} = 0.5$. Given the multiplier $\frac{1}{1 - 0.5} = 2$, the short run output $\tilde{Y}$ would increase by 2%

Multiplier Effects

- Initial Impact: Increase in investment leads to job creation among construction workers
- Consequent increase in consumption: New employed would increase their consumption
- Impact on Retailers: retailer and manufacturers are booming as demand increase, compelling them to consider hire more staff
- Further job gain: more hiring in the retail sector result in additional unemployment, causing a positive effect on workers in these industries
- Continued Consumption increase: workers further increase their consumption, creating a cycle that affects other businesses
- Chain Reaction: The initial reduction in investment triggers a prolonged chain of effects, impacting various sectors of the economy through interconnected job gain and increase consumption

Multiplier Effects



- The investment equation

$$I_t = \bar{a}_i \bar{Y}_t - \bar{b} (R_t - \bar{r}) \bar{Y}_t$$

- The two determinants of the investment at the firm level:
 - The gap between the real interest rate and the MPK
 - Cash flow
 - The amount of internal resources a company has available after paying its expenses
 - Cash flow is incorporated in the potential output term – more the economy/firm is producing, more cash is flowing in
 - We may add a term with $\tilde{Y}$ as well – another multiplier effect

- More generally, the MPK is a measure of return on capital. We may extend it by:
 - In a simple model, the return on capital = $\text{MPK} - \text{depreciation}$
 - A richer framework includes:
 - Corporate income taxes
 - Investment tax and tax credits
 - Depreciation allowances

Investment: Agency Problems

- Investment spending can be financed through: Cash flow or Borrowing
- Borrowing is typically more expensive than using internal funds – but firm with low cash flow would need to borrow to finance spending
- Agency Problems in the financial market lead to higher cost of borrowing
 - There is asymmetric information between individuals involved in a transaction – borrower may have information that the lender does not know
 - Adverse selection: when a firm knows it is in a vulnerable position, it may still borrow as the banks do not have the same information
 - Moral hazard: after the firm borrows the money, they may pick a different investment
 - Bank has to take these into account and charge a premium to cover the cost of default as a whole

Government Purchases

- Government purchases of goods and services are an important source of demand
 - In recent years, government purchases are about 20% of GDP
 - Government spending occur at all level of government, includes purchases of goods and services, transfer payments, and interest payments on debt
- Government purchases can be a source of short-run fluctuations or an instrument to reduce fluctuations
 - Election cycles also typically result in changes in government purchases
 - Government can finance different infrastructure project and increase AD – changes in $\bar{a}_g$

Fiscal Policy: Discretionary Fiscal Policy

Discretionary fiscal policy can be Government purchase or Tax Policy – The government could try and stimulate demand back to the level of potential output

- Government purchases
 - e.g., American Recovery and Reinvestment Act of 2009 (\$800B), Canada's Economic Action Plan (\$50 - 60B)
- Tax Policy
 - e.g., Investment tax credit of 1961, Economic Growth and Tax Relief Reconciliation Act of 2001, CERB

Fiscal Policy: Automatic stabilizers

- The other channel that the government can affect the economy is through *automatic stabilizers*
- Automatic stabilizers are also part of fiscal policy but do not require any additional action to implement
- Transfer spending is the main source of fiscal policy automatic stabilization (e.g., unemployment insurance, Medicare)
 - During the recession, more people would be unemployed, this implies more people would apply for Unemployment Insurance
- Progressive tax schedule would be the other automatic stabilizer
 - The tax rate is lower when the worker earn less income

Progressive Tax Schedule

- Tax schedule in year 2021 and Average tax rate by income

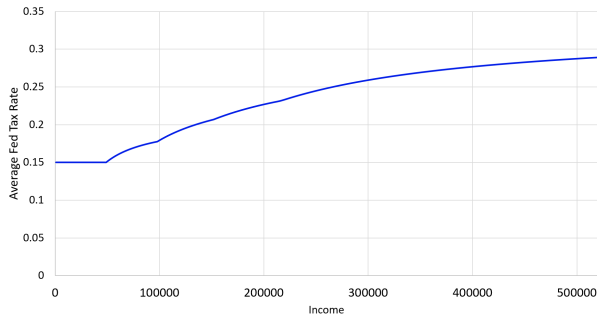
Part A – Federal tax on taxable income

Use the amount from line 26000 to complete the appropriate column below.

	Line 26000 is \$49,020 or less	Line 26000 is more than \$49,020 but not more than \$98,040	Line 26000 is more than \$98,040 but not more than \$151,978	Line 26000 is more than \$151,978 but not more than \$216,511	Line 26000 is more than \$216,511	
Amount from line 26000						67
Line 67 minus line 68 (cannot be negative)	– 0 00	– 49,020 00	– 98,040 00	– 151,978 00	– 216,511 00	68
	=	=	=	=	=	69
Line 69 multiplied by the percentage from line 70	× 15%	× 20.5%	× 26%	× 29%	× 33%	70
	=	=	=	=	=	71
Line 71 plus line 72	+ 0 00	+ 7,353 00	+ 17,402 10	+ 31,425 98	+ 50,140 55	72
Federal tax on taxable income	=	=	=	=	=	73

Enter the amount from line 73 on line 116 and continue at line 74.

Average Fed Tax Rate



- The impact of fiscal policy depends on timing and the “no free lunch” principle
- The problem of timing:
 - Discretionary changes often take a long time to implement. Tax credits, construction, education spending, and the like are put into place with delays
 - Most discretionary fiscal policy require acknowledging a change in the economy, drafting legislation, passed by parliament, and approval by the executive branch
 - Issues can arise and fiscal policy can have unintended effects if the shock that the legislation was designed to mitigate has passed

- The “no free lunch” principle:
 - Government also faces a budget constraint
 - $\text{Government spending} + \text{transfer payments} = \text{tax revenue}$
 - If it increases spending today, at some point in the future it must raise taxes or reduce spending
 - Expecting the changes in the future, it may offset the stimulus today

Ricardian Equivalence

- According to Ricardian equivalence:
 - The timing of tax changes does not matter for consumer behavior
 - Permanent income hypothesis: Consumption today is determined by the present discounted value of your lifetime income, after taxes
 - The present value of government tax collection determines behavior
- Suppose government decides to increase government purchases by \$500 million



- The government spending leads to an increase in $\bar{a}_g$ (and G), but, given the expectation of an increase in future tax, the stimulus effect may be offset by the decrease in consumption

- Trade balance = Net exports

$$EX - IM = \bar{a}_{ex} \bar{Y}_t - \bar{a}_{im} \bar{Y}_t = (\bar{a}_{ex} - \bar{a}_{im}) \bar{Y}_t$$

- When $EX > IM$, we have a trade surplus, $NX > 0$
- When $EX < IM$, we have a trade deficit, $NX < 0$
- Suppose there is an increase in demand for Canadian goods from Europe ($\bar{a}_{ex}$) $\implies$ IS shifts right $\implies$ an increase in short-run output
- Suppose there is an increase in Canadian's demand for Spanish wine ($\bar{a}_{im}$) $\implies$ IS shifts left $\implies$ a fall in short-run output

Next Lecture

- Next lecture will begin Ch.12